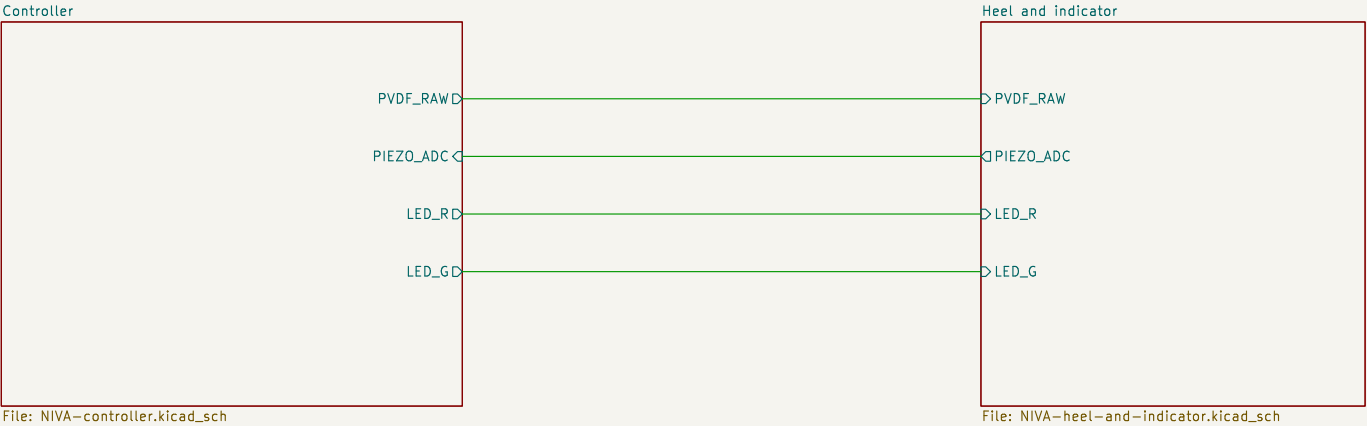


NIVA pod Rev C – one electrically connected hierarchy

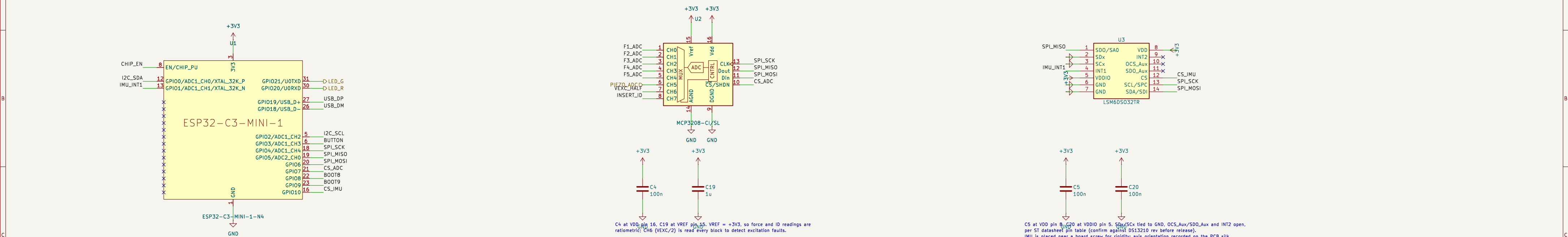


Rails +3V3, GND and VBAT are global power symbols shared by both sheets.
Cross-sheet signals travel only through the four sheet pins drawn above.
ENGINEERING PROTOTYPE – NOT FOR FABRICATION. See README for the open review items.

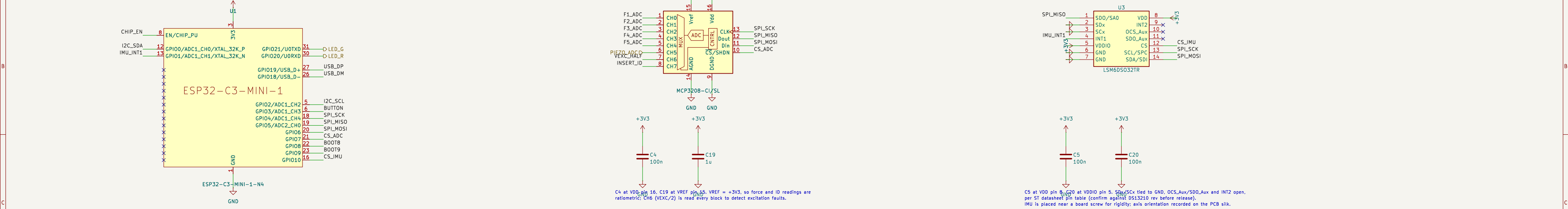
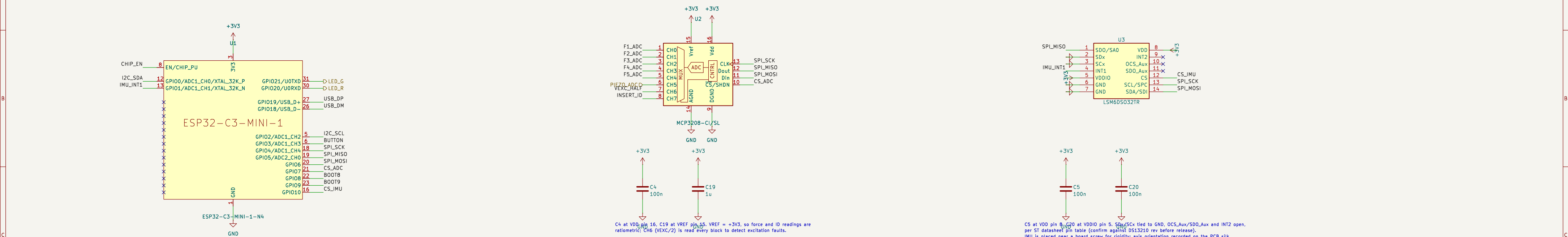
Generated by work/build_niva_schematic_revC.py NOT FOR FABRICATION. Values, protection and layout require bench validation. NIVA		
Sheet: / File: NIVA-pod.kicad_sch		
Title: NIVA pod Rev C hierarchy		
Size: A4	Date: 2026-09-28	Rev: C – ENGINEERING PROTO
KiCad E.D.A. 9.0.9		Id: 1/3

NIVA POD / CONTROLLER + ACQUISITION / REV C ENGINEERING PROTOTYPE – NOT FOR FABRICATION

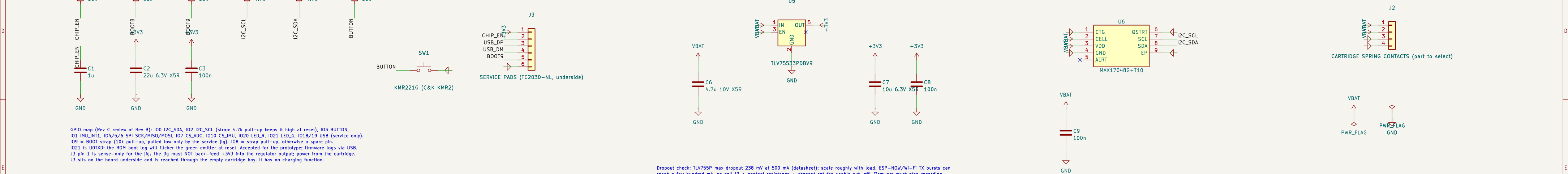
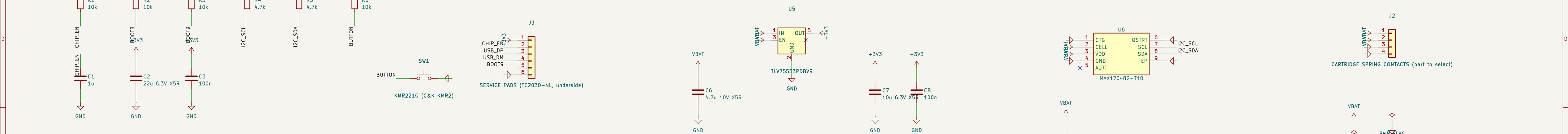
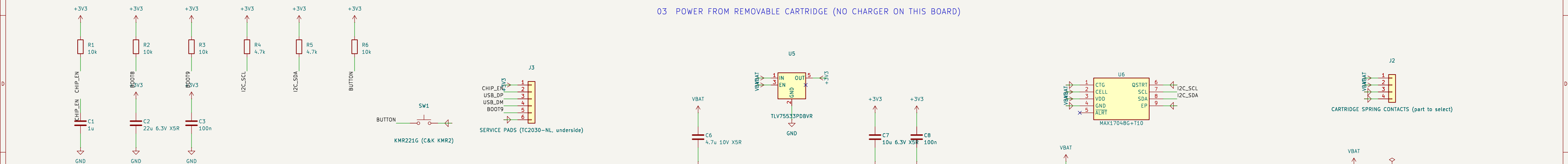
01 ESP32-C3 MODULE, BOOT STRAPS, SERVICE 02 12-BIT ADC + INERTIAL SENSOR (SHARED SPI)



01 ESP32-C3 MODULE, BOOT STRAPS, SERVICE 02 12-BIT ADC + INERTIAL SENSOR (SHARED SPI)



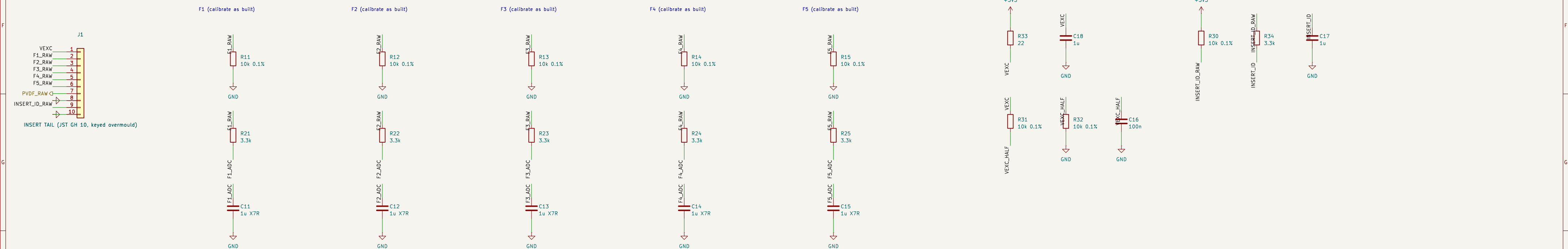
03 POWER FROM REMOVABLE CARTRIDGE (NO CHARGER ON THIS BOARD)



GPIO map (Rev C (review of Rev B)): I00 I2C_SDA, I02 I2C_SCL (strap: 4.7k pull-up keeps it high at reset), I03 BUTTON, I01 MML1NT1, I04/5/6 SPI SCK/MISO/MOSI, I07 CS_ADC, I010 CS_JMU, I020 LED_R, I021 LED_G, I018/19 USB (service only), I09 = BOOT strap (I04 pull-up, pulled low only by the service jig), I08 = strap pull-up, otherwise a spare pin, I021 is U07XD: the ROM boot log will flicker the green emitter at reset. Accepted for the prototype; firmware logs via USB, J3 pin 1 is sense-only for the jig. The jig must NOT back-feed +3V3 into the regulator output; power from the cartridge. J3 sits on the board underside and is reached through the empty cartridge bay. It has no charging function.

Dropout check: TLV755P max dropout 238 mV at 500 mA (datasheet); scale roughly with load. ESP-NOW/Wi-Fi TX bursts can reach a few hundred mA, so cell IR + contact resistance + dropout set the usable cut-off. Firmware must stop recording at a gauge-reported threshold set from measured sag (bench item PJ). C6 input/C7/C8 + C2/C3 output bulk.
J2: 2 x VBAT + 2 x GND spring contacts land on pads on the cartridge lid. Cell protection (e.g. bq27970) lives in the cartridge; charging, NTC and pack ID belong to the separate dock (not designed here).

04 PASSIVE INSERT INTERFACE / FIVE DIVIDER CHANNELS (BENCH TOPOLOGY, NOT A201 REFERENCE CIRCUIT)



Force channel: FSR from VEXC to Fin_RAW, 10k to GND, $3.3k + 1\ \mu\text{F}$ low-pass ($f_c = 48\ \text{Hz}$) to the ADC. The $1\ \mu\text{F}$ reservoir is $\sim 50,000\times$ the MCP3208 sample capacitor, so acquisition settling is set by the RC, not the ADC source-impedance limit. Excitation is continuous (Rev B decision kept): switching it every 5 ms cannot settle a 3.3 ms RC. R33 limits a tail short to $\sim 150\ \text{mA}$ so the 3V3 rail survives; VEXC is sensed after R33 (Rev B sensed +3V3, which could not detect this fault). ESD at J1: every force (p) line reaches the ADC through 3.3k into a $1\ \mu\text{F}$ reservoir ($8\ \text{M}\Omega \times 1\ \mu\text{F} \rightarrow 0.8\ \text{V-step}$). No TVS is fitted: the RC networks are the protection. System-level IEC 61000-4-2 testing is still required. Divider output is NOT the Tekscan A201 op-amp reference circuit and does not inherit its linearity or drift figures.

Force channel: FSR from VEXX to F_N_RAW, 10k Ω to GND, 3.3k Ω + 1 μ F low-pass (f_c = 48 Hz) to the ADC. The 1 μ F reservoir

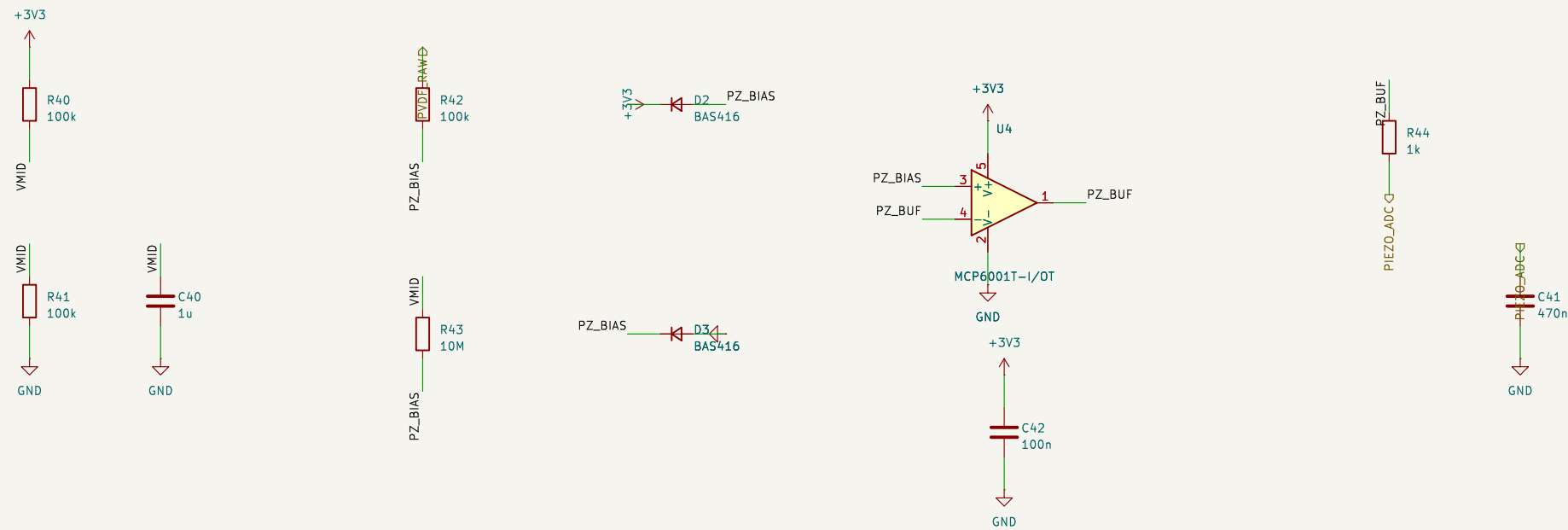
Size: A2	Date: 2026-09-28	Rev: C – ENGINEER
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to -150 mA so the 3V3 rail survives; VEXC is sensed after R33 (Rev B sensed +3V3, which could not detect this fault);
ESD at J1: every force/ID line reaches the ADC through 3.3k into a 1 uF reservoir (8 kV HBM = 0.8 uC → 0.8 V step).

No FVS is fitted; the RC networks are the protection. System-level IEC 61000-4-2 testing is still required.
Divider output is NOT the Tekscan A201 op-amp reference circuit and does not inherit its linearity or drift figures.

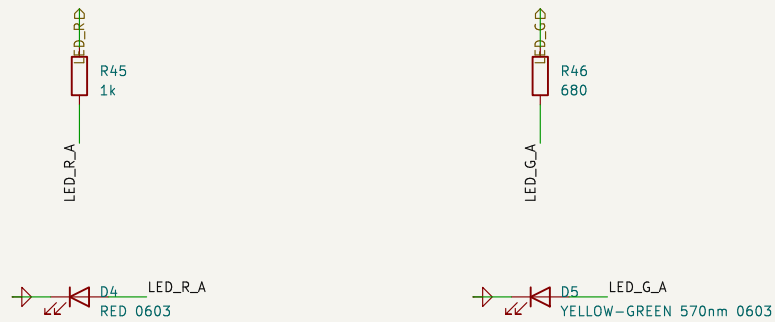
NIVA POD / HEEL PVDF BUFFER + STATUS / REV C ENGINEERING PROTOTYPE

01 MID-RAIL BIAS, INPUT PROTECTION, BUFFER



PVDF film (~1.4 nF class) with 10M bias: high-pass ~11 Hz. R42 limits clamp current during heel-strike spikes. D2/D3 clamp the buffer input to the rails. Low-leakage diodes: leakage x 10M sets the offset budget (check datasheet). R44/C41 = 339 Hz single pole for 2 kHz sampling: only ~10 dB at Nyquist. Treat as exploratory; oversample and decimate in firmware or add a 2nd-order stage after the E3 cable-motion experiment.

02 STATUS EMITTERS: RED + YELLOW-GREEN UNDER ONE LIGHT WINDOW



~1.3–1.7 mA per emitter at 3.3 V. Red + green together is read as amber. Light-window brightness and colour separation must be checked in the printed cover. Status colours signal recording quality only, never a clinical result.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.

NIVA

Sheet: /Heel and indicator/
File: NIVA-heel-and-indicator.kicad_sch

Title: NIVA pod | Heel vibration buffer and status emitters | Rev C

Size: A3 Date: 2026-09-28

KiCad E.D.A. 9.0.9

Rev: C - ENGINEERING PROTO

Id: 3/3